



Dr. Younes Bouhadjar

POSTDOC RESEARCHER

Jülich, Germany

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Work Experience

Postdoctoral Researcher

Jülich, Germany

NEUROMORPHIC SOFTWARE ECOSYSTEMS, JÜLICH RESEARCH CENTER

Since 10/2022—

- Developed a suite of **sequence processing tasks** based on artificial grammars with tunable complexity and multiple encoding schemes to investigate learning mechanisms in artificial and biological systems.
- Designed a lightweight benchmark for **state space and linear transformer models**, enabling systematic evaluation of gating, forgetting, and selectivity in sequence learning.
- Contributed to **NeuroBench**, a community-driven framework for standardized benchmarking of neuromorphic computing across algorithmic and hardware tracks.
- **Fine-tuned** large language models to promote high activation sparsity using surrogate gradient.
- Contributed actively to research on **quantization** effects in state-space models and their deployment on memristive hardware.
- Co-developed an **FPGA**-based spiking neural network implementation with axonal delays.
- Improved the **spiking TM model** (see below) to learn complex sequences.
- Supervised several master's theses on training **spiking neural networks** and the role of dendritic computing.

Doctoral Researcher

Jülich, Germany

COMPUTATIONAL AND SYSTEMS NEUROSCIENCE & ELECTRONIC MATERIALS INSTITUTES, JÜLICH RESEARCH CENTER

10/2018–10/2022

- Developed a spiking TM model for sequence learning, prediction, and generation in networks of spiking neurons
- Provided a mechanistic explanation of role of oscillation and noise in biological system
- Built neuromorphic hardware for the above sequence learning model centered around memristive devices
- Contributed to the development of neural simulator NEST and NESTML

Research Intern

San Jose, CA, USA

IBM ALMADEN RESEARCH CENTER

03/2018–09/2018

- Developed and implemented a memory-augmented neural network model inspired by the human working memory
- Implemented psychometric tests to assess the performance of the model
- Implemented machine learning models for visual question answering (VQA)
- Designed and implemented a machine learning framework: <https://github.com/IBM/mi-prometheus>

Research Intern

Yorktown Heights, NY, USA

IBM T. J. WATSON RESEARCH CENTER

06/2017–08/2017

- Developed a custom software for operating a novel optical sensor, processing the data, and applying fitting routines for noise removal

Education

Ph.D in Neuromorphic Computing & Engineering

Aachen, Germany

RWTH AACHEN UNIVERSITY

10/2018–02/2023

- Thesis: brain inspired sequence learning algorithm and foundations of a memristive hardware implementation
- Supervisors: Dr. Tom Tetzlaff & Dr. Dirk J. Wouters
- Topics: sequence learning, spiking neurons, neural plasticity, probabilistic computing, memristive devices
- Grade: Magna Cum Laude (with great distinction)

M.Sc in Micro & Nanotechnologies for Integrated Systems

Grenoble, France

PHELMA INP GRENoble

09/2016–09/2018

- Joint degree between EPFL Switzerland, PHELMA INPG France, and Politecnico di Torino Italy
- Topics: analog and digital circuit design, semiconductor devices, nanoscience, microelectromechanical system design
- Grade: 16.18/20 (very good)

B.Sc in Physics and Electronics

Grenoble, France

PHELMA INP GRENoble

09/2013–09/2016

- Topics: classical and modern physics, linear algebra, differential equations, probability theory
- Grade: 16.65/20 (very good)

Selected Publications

Proceedings

- 2025 **Bouhadjar, Y.**, Fabre, M., Schmidt, F., and Neftci, E. (2025). Dissecting linear recurrent models: how different gating strategies drive selectivity and generalization. Under review at AAAI.
- 2025 Zhao, Y., **Bouhadjar, Y.**, and Neftci, E. (2025). Efficient FPGA Implementation of Spiking Neural Networks with Integrated Axonal Delays.
- 2025 **Bouhadjar, Y.***, Lober, M*, Neftci E., Diesmann, M., Tetzlaff, T. (2025). Unsupervised continual learning of complex sequences in spiking neuronal networks. Accepted to be published in Proceedings of the International Conference on Neuromorphic Systems.
- 2025 Siegel, S., Yang, M., **Bouhadjar, Y.**, Fabre M., Neftci E., and Strachan, J. P. (2025) QS4D: Quantization-aware training for efficient hardware deployment of structured state-space sequential models memory.
- 2024 Feiler, F., Neftci, E., and **Bouhadjar, Y.** (2024). Unsupervised Learning of Spatio-Temporal Patterns in Spiking Neuronal Networks. In 2024 International Conference on Neuromorphic Systems (ICONS) (pp. 366-370).
- Siegel, S., Ziegler, T., **Bouhadjar, Y.**, Tetzlaff, T., Waser, R., Dittmann, R., and Wouters, D. (2023).
- 2023 Demonstration of neuromorphic sequence learning on a memristive array. In Proceedings of the 2023 Annual Neuro-Inspired Computational Elements Conference (pp. 108-114).
- 2023 Benmeziane, H., Ounnoughene, A. Z., Hamzaoui, I., and **Bouhadjar, Y.** (2023). Skip Connections in Spiking Neural Networks: An Analysis of Their Effect on Network Training. arXiv preprint arXiv:2303.13563.
- Bouhadjar, Y.***, Caterina Moruzzi*, Melika Payvand*. (2022). Prediction: An Algorithmic Principle Meeting Neuroscience and Machine Learning Halfway. In Proceedings of the 3rd International Workshop on Human-Like Computing at the 2nd International Joint Conference on Learning & Reasoning (pp. 1-7).
- Bouhadjar, Y.**, Diesmann, M., Wouters, D. J., and Tetzlaff, T. (2020). The speed of sequence processing in biological neuronal networks. In Proceedings of the Neuro-inspired Computational Elements Workshop (pp. 1-2).
- Bouhadjar, Y.**, Diesmann, M., Waser, R., Wouters, D. J., and Tetzlaff, T. (2019). Constraints on sequence processing speed in biological neuronal networks. In Proceedings of the International Conference on Neuromorphic Systems (pp. 1-9).
- Jayram, T. S.*, **Bouhadjar, Y.***, McAvoy, R. L., Kornuta, T., Asseman, A., Rocki, K., and Ozcan, A. S. (2018). Learning to remember, forget and ignore using attention control in memory. ArXiv:1809.11087.
- 2018 (* shared first author)

Journals

- 2025 Yik, J., Van den Berghe, K., den Blanken, D., **Bouhadjar, Y.**, ... and Reddi, V. J. (2025). The neurobench framework for benchmarking neuromorphic computing algorithms and systems. Nature Communications, 16(1), 1545.
- 2023 **Bouhadjar, Y.**, Siegel, S., Tetzlaff, T., Diesmann, M., Waser, R., and Wouters, D. J. (2023). Sequence learning in a spiking neuronal network with memristive synapses. Neuromorphic Computing and Engineering, 3(3), 034014.
- 2023 **Bouhadjar, Y.**, Wouters, D. J., Diesmann, M., and Tetzlaff, T. (2023). Coherent noise enables probabilistic sequence replay in spiking neuronal networks. PLOS Computational Biology, 19(5), e1010989.
- Siegel, S., **Bouhadjar, Y.**, Tetzlaff, T., Waser, R., Dittmann, R., and Wouters, D. J. (2023). System model of neuromorphic sequence learning on a memristive crossbar array. Neuromorphic Computing and Engineering, 3(2), 024002.
- 2022 Oberländer, J, **Bouhadjar, Y.**, and Morrison, A. (2022). Learning and Replaying Spatiotemporal Sequences: A Replication Study. Frontiers in integrative neuroscience, 113.
- 2022 **Bouhadjar, Y.**, Wouters, D. J., Diesmann, M., and Tetzlaff, T. (2022). Sequence learning, prediction, and replay in networks of spiking neurons. PLOS Computational Biology, 18(6), e1010233.

Software

- 2023 NESTML 8.0.0 (<https://github.com/nest/nestml/releases/tag/v8.0.0>)
- 2023 NESTML 4.0.0 (<https://github.com/nest/nestml/releases/tag/v4.0>)
- 2023 NEST 3.0 (<https://zenodo.org/records/4739103>)

In preparation

- Zajzon, B., **Bouhadjar, Y.**, Fabre, M., Schmidt, F., Ostendorf, N., Neftci, E., Morrison, A., and Duarte, R. (2025). SymSeqBench: a unified framework for the generation and analysis of rule-based symbolic sequences and datasets.
- 2025 Zajzon, B., **Bouhadjar, Y.**, Tetzlaff, T., Morrison, A., Duarte R. (2025). Sequence learning under biophysical constraints: a re-evaluation of prominent models.
- 2025 Lober, M., **Bouhadjar, Y.**, Diesmann, M., Tetzlaff, T. (2025). Learning sequence timing and control of recall speed in networks of spiking neurons.

Presentations

Talks

- 2023 **Bio-inspired sequence learning mechanisms and their implementation in a memristive neuromorphic hardware** *Hanover, Germany*
NNPC conference
- 2022 **The neuronal mechanisms of sequence processing in biological networks** *Online*
AI Summit, Algiers
- 2021 **Sequence learning, prediction, and generation in networks of spiking neurons** *Online*
NEST conference (lightning talk)
- 2021 **Sequence learning, prediction, and generation in networks of spiking neurons** *Online*
Annual Neuro-Inspired Computational Elements (NICE)
- 2019 **Constraints on sequence processing speed in biological neuronal networks** *Knoxville, United States*
International Conference on Neuromorphic Systems (ICONS)
- 2019 **Constraints on sequence processing speed in biological neuronal networks** *Jülich, Germany*
INM-ICS retreat

Posters

- 2024 **Training Spiking Neural Networks to emulate brain-like activity for optimal efficiency** *La Jolla, United States*
International Conference on Neuromorphic Computing and Engineering
- 2023 **Probabilistic sequential memory recall in spiking neuronal networks** *Leipzig, Germany*
Computational Neuroscience Meeting
- 2022 **Sequence learning in a spiking neural network with memristive synapses** *Groningen, Netherlands*
Materials, devices and systems for neuromorphic computing (MatNeC), **best poster prize**
- 2021 **Sequence learning, prediction, and generation in networks of spiking neurons** *Online*
Annual Computational Neuroscience meeting (CNS)
- 2021 **Sequence learning, prediction, and generation in networks of spiking neurons** *Online*
Annual Neuro-Inspired Computational Elements (NICE)
- 2019 **Constraints on sequence processing speed in biological neuronal networks** *Berlin, Germany*
Bernstein conference
- 2019 **Constraints on sequence processing speed in biological neuronal networks** *Knoxville, United States*
International Conference on Neuromorphic Systems (ICONS)
- 2019 **Constraints on sequence processing speed in biological neuronal networks** *Jülich, Germany*
INM-ICS retreat

Teaching Experience

Tutor: Introduction to Computational Neuroscience

RWTH, AACHEN

- Neuron models
- Probabilistic description of neuronal signals

Aachen
01/2018–05/2022

Tutor: Theoretical Neuroscience: Correlation structure of neuronal networks

RWTH, AACHEN

- Measures of pairwise correlation
- Correlations in linear systems
- Decorrelation of neural-network activity by inhibitory feedback

Aachen
01/2018–05/2022

Student Supervision

Seghairi Abderraouf

JÜLICH RESEARCH CENTER

- Thesis: Fine-tuning large language models to promote high activation sparsity

Jülich, Germany

10/2025–

Can Kayser

JÜLICH RESEARCH CENTER

- Thesis: Dendritic subunits to bridge timescales in spiking neuronal network learning

Jülich, Germany

10/2025–

Tarun Phanindra Maddu

JÜLICH RESEARCH CENTER

- Thesis: Learning sequences of lip reading using state space models

Jülich, Germany

10/2024–4/2025

Felix Schmidt

JÜLICH RESEARCH CENTER

- HiWi: enhancing efficiency in neural networks through the incorporation of dendritic computing

Jülich, Germany

10/2023–3/2025

Research Summer Camp

SCHOOL AI ALGIERS, MENTORING 4 MASTER STUDENTS

- Thesis: Towards an understanding of computation through time encoding in spiking neural networks

Jülich, Germany

06/2023–09/2023

Florian Feiler (Master thesis)

JÜLICH RESEARCH CENTER

- Thesis: Online Unsupervised Learning in Brain-Inspired Neural Networks

Jülich, Germany

05/2023–11/2023

Jette Oberländer (Bachelor thesis)

JÜLICH RESEARCH CENTER

- Thesis: Learning and Replaying Spatiotemporal Sequences: A Replication Study

Jülich, Germany

11/2021–09/2022

Hubertus Borsch (Master thesis)

JÜLICH RESEARCH CENTER

- Thesis: Learning spatiotemporal sequences with spiking neural networks

Jülich, Germany

04/2021–04/2022

Personal Skills

MATHEMATICS

- Probability theory
- Linear algebra
- Non-linear systems
- Differential/integral calculus

PROGRAMMING

- Python, Matlab, C, C++, CUDA, Triton

SCIENTIFIC COMPUTING

- Simulation, data analysis and visualization with Python
- Modeling and simulation of spiking neural networks in NEST
- Training and inference of neural networks in PyTorch
- Open source development using GitHub
- Linux (Debian)

TOOLS

- Git, Github, Docker

HARDWARE DESIGN

- Verilog, VHDL

Service & Outreach

ICONS Conference Program Committee

ARLINGTON

- Reviewed papers

Virginia, USA

05/2024–08/2024

ICONS Conference Program Committee

SANTA FE

- Reviewed papers

NM, USA

05/2023–08/2023

Doctoral representative

JÜLICH RESEARCH CENTER

- Worked on improving the working conditions of doctoral researchers
- General committee work and representative tasks

Jülich, Germany

01/2020–12/2020

Helmholtz Junior representative

JÜLICH RESEARCH CENTER

- Enhance networking and share best practices
- Helped organize a mental health awareness month

Jülich, Germany

01/2020–12/2020

Content Curation

JÜLICH RESEARCH CENTER

- Managing IT infrastructure
- Support in implementing reproducible research

Jülich, Germany

01/2021–